1.5x the width does not buy the MLP back

N=48, degree 5, 4 colours, leak-free | stage C closed 15/15, stage D's width arm at 3/3 seeds | 72 cells

THE TEST NO EARLIER CELL IN THIS GRID PERFORMED

Every ablated rung measured before stage D ran at AT MOST the hub's depth and at most its width, so no ablated model had ever been handed MORE of a resource than the model it is scored against. D2 is attention-only -- no MLP anywhere -- at d_model 384, 1.5x the hub's residual stream. At 3,799,354 trainable against the hub's 4,899,898 it is still the smaller model; what it has more of is WIDTH.

WHAT THE PRE-REGISTERED SCORER RETURNS

rho_50 median 0.760 against the hub's 0.592, mean_solve 0.3155 [spread 0.0075] against 0.5389, 3 of 3 seeds crossing, plateau escapes 3/3. Attention-only at the hub's own width sits at 0.763 / 0.3039. D2's registered prediction was rho_50 <= 0.740 -- that 1.5x width buys the MLP back. The scorer's call on it is HELD, printed in full on page 6.

AND THE CALL WAS ON RECORD BEFORE THE CELL EXISTED

Stage E's header, committed while stage D was still running, explicitly backed D2's FALSIFIER over D2's own prediction: if the barrier reading is right then capacity in any form is not the currency, so d384 attention-only should land beside d256 attention-only rather than reaching the hub. That is a registered prediction about another job's unfinished cell, and it is scored here exactly as it was written.

rho_50 = realized hint density at which solve rate crosses 0.5 on a 14-point x 400-instance grid, LOWER IS BETTER, reported CENSORED when no bin crosses. mean_solve = unweighted mean over the 14 bins, which never censors. The seed-spread bar s* = 0.1484 / 0.2918 was frozen before any of these cells ran. Commit 208d2a5.

What the model actually sees

Every token is one categorical id into a 58-symbol vocabulary (N=48, q=4, degree=5). No text, no digits-as-characters.

VOCABULARY

0-5	BEGIN_HINTS BEGIN_GRAPH BEGIN_REASONING
	BEGIN_ROUND SOLVED UNRESOLVED
	markers (UNRESOLVED = DD got stuck; unused here)
6-9	COLOR_0 COLOR_1 COLOR_2 COLOR_3
	the q colours (token name COLOR_n, prose 'colour')
10-57	N0 N1 ... N47
	one id per node, no structure in the id

GRAMMAR

BEGIN_HINTS	(NODE COLOR)*
BEGIN_GRAPH	(NODE (NODE){degree})*
BEGIN_REASONING	(BEGIN_ROUND (NODE COLOR)+)*

SOLVED

*regex: * = zero or more, + = one or more, {degree} = exactly 5*

The graph section is a neighbour table, not an image: node, then its 5 neighbours.

THE ORDER WITHIN A ROW IS THE LEAK.

THIS INSTANCE, 401 TOKENS

HINTS	BEGIN_HINTS N0 COLOR_0 N1 COLOR_2 N2 COLOR_3 N10 COLOR_1 N15 COLOR_1 N17 COLOR_3 ... (37 tokens)
GRAPH	BEGIN_GRAPH N0 N16 N8 N38 N34 N17 N1 N13 N27 N34 N28 N10 ... (289 tokens)
REASONING	BEGIN_REASONING BEGIN_ROUND N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 N34 COLOR_1 N44 ... (75 tokens)

TRAINING LOSS is next-token cross-entropy on EVERY token, the problem statement included -- there is no section mask.
AT EVAL the model is PROMPTED with everything through BEGIN_REASONING and generates the rest: 13 rounds, 30 deductions,
60 node+colour tokens, plus the BEGIN_ROUND markers and the SOLVED/UNRESOLVED terminator, which are trained targets like any other token.

DD is single-candidate elimination -- sudoku's 'naked single' rule transplanted to graph colouring, which is where the code's sudoku names come from; the task itself is the graph, not a 9x9 grid. Instance shown: seed 50000000018, FALL-colouring ensemble (every node sees all q-1 other colours, so no node is permanently undeducible).

One real reasoning trace

18 hints of 48 nodes, then 13 rounds of simultaneous deduction. Hints: N0=COLOR_0 N1=COLOR_2 N2=COLOR_3 N10=COLOR_1 N15=COLOR_1 N17=COLOR_3 ...

THE TRACE, AS EMITTED

```
BEGIN_ROUND        N9 COLOR_3 N14 COLOR_2 N16 COLOR_2 N32 COLOR_0 ...
  6 deduced
BEGIN_ROUND        N42 COLOR_3 N45 COLOR_1
  2 deduced
BEGIN_ROUND        N13 COLOR_0 N20 COLOR_0 N37 COLOR_2
  3 deduced
BEGIN_ROUND        N21 COLOR_2 N24 COLOR_3 N31 COLOR_2
  3 deduced
... 9 further rounds

SOLVED
```

WHY ONE DEDUCTION FIRES

node N9, neighbours N25 N10 N29 N31 N18

their colours known before this round:

```
[0, 1, 2]  (3 = q-1 of the 4)

(3 known neighbours, 3 DISTINCT colours)

still unknown: N25 N31
```

only colour left: 3

```
so the trace emits  N9 COLOR_3
```

The model is never shown this reasoning -- only the node/colour pair.

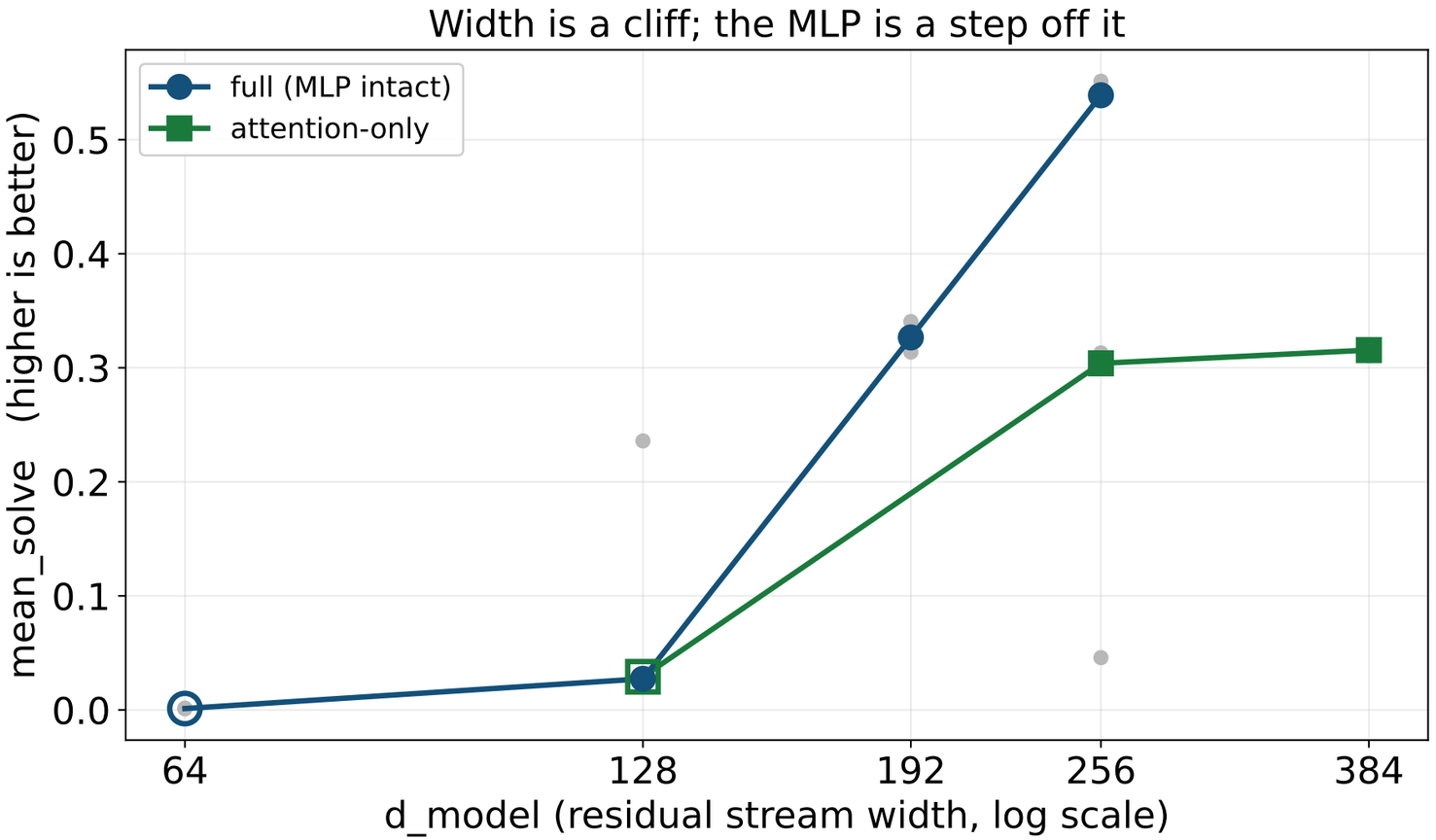
A round deduces every node whose neighbours already cover q-1 colours, all at once. Reasoning depth T is the number of rounds: a deduction in round 3 depends on ones made in rounds 1 and 2, which is the serial computation that network depth has to supply.

N=48, degree 5, 4 colours, --shuffle-adj --fall: the task every cell in this deck is trained and scored on. The attention-only models have to produce this with no MLP anywhere in the stack.

T=13 rounds, the 90th percentile of 40 draws from this config (range 1-16) -- one instance, shown to make the task legible; no result in this deck is computed from it. The worked deduction on the right uses the state BEFORE this round, which is what DD itself saw; it is recomputed from the graph, not read out of the model.

Two width ladders: the complete model, and the same model with no MLP

L=6, 4 heads, ff=1024 wherever an MLP exists; everything else at the hub's settings, 3 seeds per rung



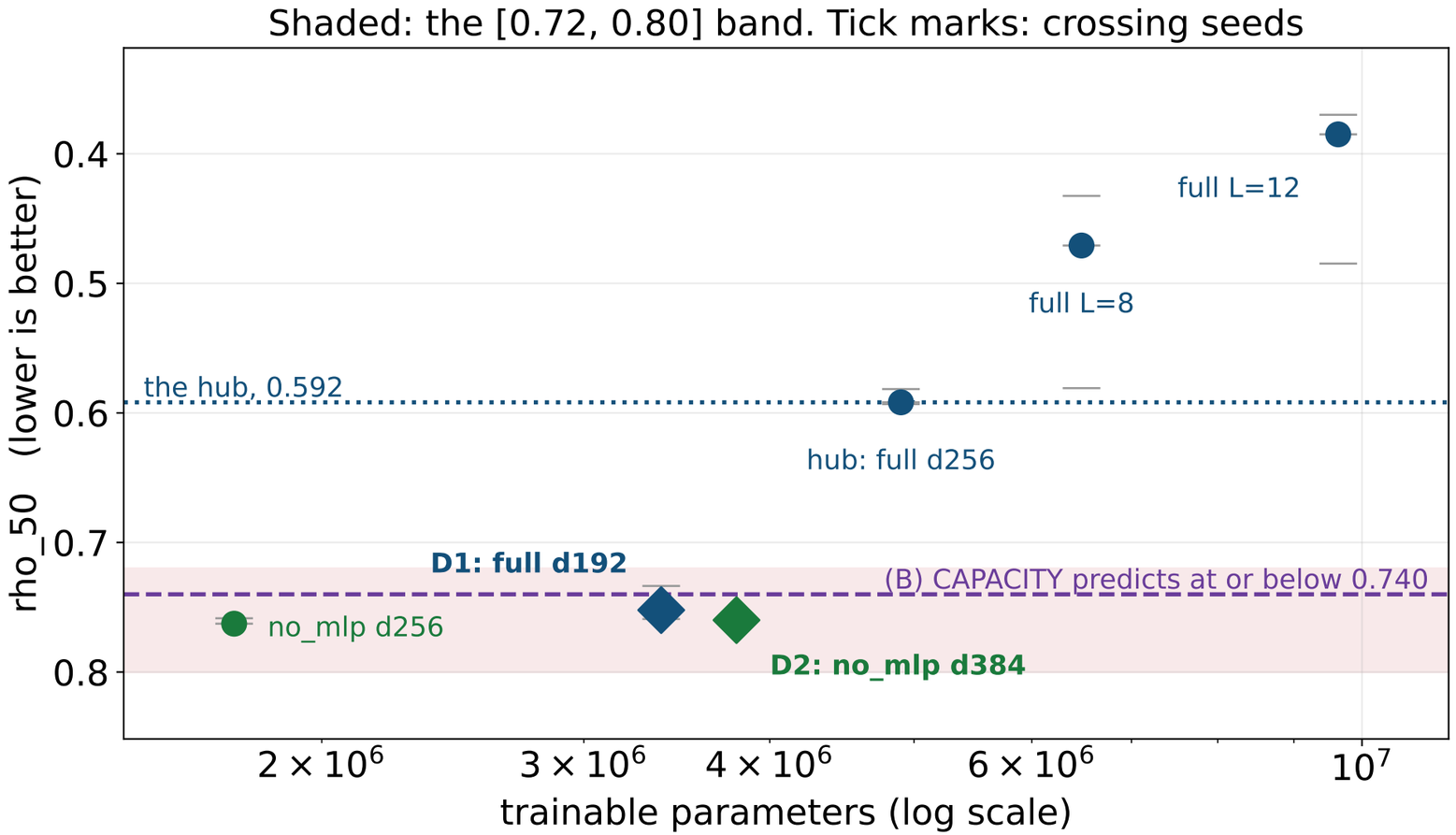
HOLLOW MARKER = never escaped the 1.64 plateau, so the cell is an optimization failure and its number is not a statement about width.

	d	mean_solve	trainable	esc
full				
	256	0.5389	4,899,898	3/3
	192	0.3266	3,381,562	3/3
	128	0.0273	2,059,834	2/3
	64	0.0013	934,714	0/3
no_mlp				
	384	0.3155	3,799,354	3/3
	256	0.3039	1,746,490	2/3
	128	0.0291	480,058	0/3

Per-seed values in grey. Per trainable parameter, width buys more than the MLP does: attention-only at d256 has fewer parameters than the complete model at d128 and solves, while the complete model at d128 never crosses on any seed. The d64 rung is drawn hollow because all three of its seeds finished ABOVE the plateau -- it did not train in 375k steps, which is a different claim from 'too narrow to deduce'. Commit 208d2a5.

Where the two stage D rungs landed, against predictions made first

Every prediction below was written into a committed sbatch header before the cell it describes finished




```
REGISTERED, THEN SCORED

D1   full d192, 3,381,562
  asked  mean_solve in [0.10, 0.45],
         and strictly between its two
         neighbours on the ladder
  got    0.3266, rho_50 0.752
         -> INSIDE the window, and
         between 0.0273 and 0.5389

D2   no_mlp d384, 3,799,354
  asked  rho_50 <= 0.740
  got    0.760

Censored rungs cannot be placed on
this axis and are left out; all of
them sit BELOW the band, so leaving
them out understates the pile-up
rather than flattering it.
```

The hub and the complete model at L=8 and L=12 are the paired controls: same 14-point grid, same 375k-step budget, same eval. s* = 0.1484 is the seed-to-seed spread measured at fixed architecture, which is why the band is a band and not a line. Commit 208d2a5.

The rule, and the one place it contradicts itself

Transcribed from the sbatch headers into scripts/capacity_back.py, pinned by 15 tests before the cells landed

D1 -- HELD | D2 -- HELD

D1: (A) HOLDS ON THIS RUNG -- median rho_50 0.752 sits inside the registered [0.72, 0.80] window, on the 0.76 rung, with more of a resource than the hub and nothing to show for it. D2: (A) HOLDS ON THIS RUNG -- median rho_50 0.760 sits inside the registered [0.72, 0.80] window, on the 0.76 rung, with more of a resource than the hub and nothing to show for it.

THE ORDER THE BRANCHES FIRE IN, WHICH IS ITSELF REGISTERED

Trainability first: 0 plateau escapes means 'did not train in 375k steps' and is NEVER evidence in either direction -- the trap stage C's P2 fell into. Then fewer than three seeds, where nothing can be scored. Then fewer than two crossing seeds, where the falsifier cannot fire. Then a median below 0.72, which ends the barrier reading on that rung alone. Then the tie-break region up to 0.740, then the band's upper edge at 0.80. A rung that never got the chance to break the reading is not counted as one that held it.

AND THE CONTRADICTION, REPORTED RATHER THAN RESOLVED QUIETLY

The stage E header states a tie-break sending any median in [0.72, 0.740] to the barrier reading, and nine lines later a falsifier killing that same reading for any median below 0.740 on 2 of 3 crossing seeds. They disagree on the whole interval [0.72, 0.740). The scorer follows the TIE-BREAK -- written for exactly that overlap, and the conservative choice, since the capacity reading is the claim needing evidence -- and PRINTS BOTH readings whenever a median lands there. Choosing one silently is how a pre-registration becomes a post-hoc one.

tests/test_capacity_back.py pins every threshold to the header both as literal text and as the arithmetic that produced it (hub + s* = 0.740, full L=8 + s* = 0.619), and pins the branch order. Moving one threshold by 0.02 turns three quarters of the tie-break region into a falsification, and nothing else in the repo would notice. Commit 208d2a5.

What this does not settle, and what is still running

Stage E (51171824) hands the same ablation one more DEPTH instead of one more width -- the second axis

THE DEPTH ARM, DELIBERATELY NOT IN THIS DECK

Attention-only at L=8 and L=12, three seeds each, everything else at the hub's settings. Both depths were chosen because the complete model is already measured there, so each rung has a paired partner at identical depth. Registered falsifier: either rung with a median below the tie-break floor on at least 2 of 3 crossing seeds ends the barrier reading, with no partial credit and no 'but the other rung held'.

THE WAY IT PRODUCES NOTHING, NAMED IN ADVANCE

Trainability, not capacity. Attention-only escapes the 1.64 plateau on 2 of 3 seeds at L=6 and 0 of 3 at L=4, so a DEEPER attention-only stack failing to escape is the expected failure mode and will be written 'did not train in 375k steps'. The budget is deliberately not raised to rescue it: a rung trained on a different budget cannot be read against the ladder it is compared to.

NO IMPROVEMENT IS CLAIMED HERE, AND NONE WAS SOUGHT

Previous best on this eval, same grid and same budget: the complete model at L=12, rho_50 0.385 / mean_solve 0.8163 at 9,638,458 trainable. Nothing in this deck approaches it. Since 11-04 the brief is the SMALLEST architecture that still learns direct deduction, so the number that matters is the standing minimum: attention-only at d256, 1,746,490 trainable, rho_50 0.763 -- 2.81x fewer parameters than the hub and 1.2 s* worse. Next, once the depth arm lands: a 3-seed replicate of whichever rung survives at N=96.

SCAN_E is absent from SCAN_DIRS and from this deck on purpose: stage E's rungs carry entirely default hyperparameters and would pool into the s* every verdict here is scored against. When it lands it gets its own write-up, not a silent edit to this one. Commit 208d2a5.